

Zishuo ZHAO

PERSONAL INFORMATION

ADDRESS: 30 Shuangqing Road, Haidian District, 100084, Beijing, China
PHONE: +86 152 1003 1562
EMAIL: zzs21(at)mails.tsinghua.edu.cn
zishuozhao0602(at)gmail.com

EDUCATION

2021 - present Ph.D student in MATHEMATICS, **Tsinghua University**, China
Advisor: Zhengwei LIU

2017 - 2020 Undergraduate Degree in MATHEMATICS and STATISTICAL SCIENCE
University of California at Santa Barbara, U.S.

TEACHING EXPERIENCE

Fall 2023; **20420124-1: Advanced Linear Algebra (1)**, TA, Prof. Bangming Deng, Tsinghua University.

Spring 2023; **30420105-3: Advanced Calculus (2)**, TA, Profs. Wenming Zou, Tsinghua University.

Fall 2022; **10421324-11: Linear Algebra**, TA, Profs. Dingxin Zhang, Tsinghua University.

Spring 2022; **10421065-4: Calculus A(2)**, TA, Profs. Jiayan Yao, Tsinghua University.

Fall 2021; **10421324-2: Linear Algebra**, TA, Profs. Jie Du, Tsinghua University.

SCHOLARSHIP

- 2023 Comprehensive Scholarship at Tsinghua University

RESEARCH INTERESTS

My current research interests are centered around operator algebras, in particular subfactor theory. I am interested in their interactions with the study of quantum information and the classification of phases of matter, especially topological orders.

PREPRINTS

Zhengwei Liu and Zishuo Zhao, Topological Orders from Reflection Positive Frustration-free Hamiltonians. arXiv:2510.20662

Jinsong Wu and Zishuo Zhao, Bimodule Quantum Markov Semigroups. arXiv:2504.09576

Zishuo Zhao, Relative Entropy for Quantum Channels. arXiv:2312.16576

Zishuo Zhao, Rios, M.A., Avetisyan, Z., Berlow, K. et al. Almost Abelian Lie groups, subgroups and quotients. arXiv:2004.04369

PUBLICATIONS

Z. Zhao, Rios, M.A., Avetisyan, Z., Berlow, K. et al. ALMOST ABELIAN LIE GROUPS, SUBGROUPS AND QUOTIENTS. J Math Sci 266, 42–65 (2022)

CONFERENCE TALKS

2025

- October Great Plains Operator Theory Symposium 2025,
University of Puerto Rico, USA
- August Quantum Theory and Operator Theory,
Beijing Institute of Mathematical Sciences and Applications, China

2024

- January Mathematical Aspects of Quantum Theory,
Tsinghua Sanya International Mathematics Forum, China

SEMINAR TALKS

2025

- June BIMSA-Tsinghua Quantum Symmetry Seminar,
Beijing Institute of Mathematical Sciences and Applications, China

2023

- December BIMSA-Tsinghua Quantum Symmetry Seminar, Beijing, China
- September Hebei Normal University, Shijiazhuang, China